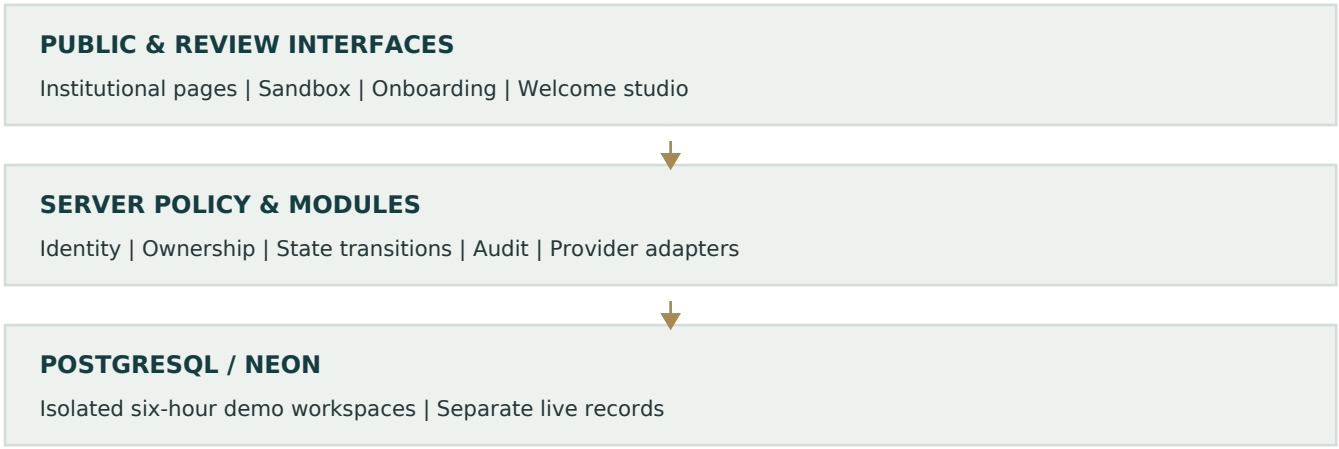
Technical architecture and demo access

Implementation, boundaries and next decisions

Current stack: Next.js 16, React 19, TypeScript, Node.js 24 and PostgreSQL/Neon. The source is hosted on GitHub and the application on Vercel. This is a working review package, not a trained STANAIC language model or a production-service assurance claim.



How the modules fit together

Boundary	Current implementation
Public content	Static institutional documents behind clean routes. Released catalogue records are read through server APIs. External source links preserve provenance.
Operational state	Role-scoped records and actions with transactional state changes. Initial platform state uses a bounded JSONB workspace record; credentials and private files use separate tables.
Existing onboarding	Single-use invitations, consent/profile/review and staff permissions remain separate from institutional designations and new operational roles.
Private files and notifications	Versioned private PDF/text downloads; protected notification outbox/preview processor. External sending and scheduler setup are separate.
External adapters	Payment checkout/webhooks, email delivery, LinkedIn publishing and optional model assistance have explicit configuration and access requirements. Missing setup fails closed.

The existing partner learning platform remains independent. Initial integration uses links; catalogue, SSO, enrolment and progress automation require agreed interfaces and access.

Roles and handoffs

A small team may hold several assignments; permissions and decision ownership remain explicit.

Operational role	Owns	Boundary
Super administrator	Staff assignments, integration setup and recovery	Membership title alone grants no software access.
Secretariat administrator	Invitations, approved designations and membership decisions	Does not own editorial or assessment outcomes.
Membership maintainer	Completeness review and recommendation	Cannot approve an appointment or grant staff access.
Course administrator	Course content, versions, publication and fee configuration	Enrolled versions cannot be silently rewritten.
Assessor	Independent review of passing assessment evidence	Separate from payment confirmation and registrar issue.
Credential registrar	Eligible sample issue, registry and revocation	Payment alone is not completion evidence.
Managing editor	Screening, reviewer assignment, decisions and publication	Author proof/licence approval precedes release.
Peer reviewer	Assigned anonymised manuscripts and confidential review	Cannot see unrelated submissions or author identity.
Finance officer	Orders, reconciliation and approved refunds	Cannot alter assessment results or issue awards.
Content editor	Approved public resource metadata	Institutional claims require authorised factual review.
Read-only auditor	Purpose-limited oversight and activity review	No state-changing operations.
Member / learner	Own profile, orders, lessons and assessment	No staff or institutional designation allocation.
Research author	Own manuscripts, revisions and proofs	No editorial decisions or reviewer identities.

Known review item: the current author proof button does not display the configured publication licence before recording consent. Add visible terms and explicit acknowledgement before real editorial use.

Access without sharing real secrets

Public demo access and private deployment configuration are different.

Demo entry	Access
Main operational sandbox	/workspace?mode=demo No personal sign-in required. Role selection affects only the isolated sandbox.
Onboarding administrator	Username: demo-admin Password: AdminDemo2026!
Onboarding member	Username: demo-member Password: MemberDemo2026!
Announcement studio	/announcements?demo=1 Fictional export-only mode; sample publication is disabled.

The two test passwords are intentionally public, restricted to synthetic onboarding workspaces, and do not authenticate against live institutional records. Workspaces expire after six hours. Do not reuse these passwords for any real account.

Credential categories already involved

The deployed backend uses server-side database connection/password variables, NEON_AUTH_COOKIE_SECRET and MAIL_ENCRYPTION_KEY. Authentication endpoints and owner/notification email settings also exist. The hosting review found these variables marked sensitive or encrypted. No values are included here; the demo link does not require sharing them.

Not connected for this review

Environment-name inventory did not show configured Stripe keys, LinkedIn access token, Resend key, dedicated live credential-signing key, notification cron secret or AI Gateway model/key. The public guide works without a model call. A provider-backed service must be configured and independently tested before it is represented as operational.

Verification and limits

Recorded audit evidence: 63 automated tests; 141 post-fix preview API checks; 36 production page/redirect cases; 37 production attachment-reset checks. All passed within their stated scope. Browser review covered filters, navigation, authoring controls, welcome exports and sample editorial/learning actions. Real OTP, charges, email delivery, LinkedIn posting/tagging, live signed issuance and model generation were not exercised. External source timeouts and automated-access denials remain recorded.

Security-review scope. A pattern scan of 122 tracked files found no recognised private-key, provider-key or password-bearing database-URL patterns. This does not establish that no secret could exist in history, external accounts or every possible deployment path. Personal attribution was removed from the public application and these handouts; repository hosting still identifies its owner.